Workshop guide: Python for Energy System Modelling

This guide sets out what each group does at each stage of the workshop, defines the key terms, and explains where to find help. The **Learning resources** document provides further reading.

Workshop materials: the workshop notebook, data and documents are available in the GitHub repository: github.com/ammarahmedwafad/pypsa-energy-modelling-workshop/tree/main/student (QR code on the slides).

Today at a glance

Working in groups, you will use a ready-made Python model in a Jupyter notebook to address two questions that energy engineers routinely face.

This is a modelling and analysis workshop, not a programming course. The code is written for you. Your job is to run the model, read the results, test what changes them and explain why, as you would in an energy consultancy or a research project. You only complete three short lines of code, and hints are provided.

Time	Activity	Question
00 to 15	Introduction	How are energy systems modelled, and why use Python and PyPSA?
15 to 20	Set-up	Open the notebook and run sections 1 to 3
20 to 35	Task 1	How much of a home's electricity can its own solar panels and a battery provide?
35 to 50	Task 2	What does a carbon price do to electricity prices and emissions?
50 to 60	Debrief, quiz and feedback	How did the groups' results differ, and why?

Why this matters. The same methods, at a much larger scale, are used by researchers, grid operators and consultancies to plan the energy transition. PyPSA, the tool in Task 2, is open source and also underpins full models of the European energy system.

By the end you will be able to:

1. load and check a year of hourly energy data in Python;
2. build a simple hour-by-hour battery model and work out how much of a home's demand its own solar can meet;
3. explain why a real battery achieves less than an ideal one;
4. build and solve a small power system model in PyPSA;
5. explain how a carbon price changes which power stations run, the electricity price and emissions.

How your group works

Each group has three or four members. Each member has a defined role and works on one screen, and the group discusses every result before it is recorded.

Role	Screen	Responsibility
Modeller	The notebook	Runs the cells and completes the code. Describes aloud what each output shows.
Reviewer	This guide and the Learning resources document	Reads each step aloud, explains the key terms, and checks the method against the resources when the group is unsure.
Reporter	The Google Form (QR code on the slide)	Verifies that every number is sensible and has the right unit, then records the group's agreed answer.
Second reviewer, in a group of four	Any	Supports the Reviewer and leads the group discussion.

Each step follows the same sequence:

1. The Reviewer reads the step aloud.
2. If the step asks for a **prediction**, agree on one as a group and the Reporter records it **before** the Modeller runs anything.
3. The Modeller runs the cell.
4. Discuss the result: does it match the prediction? Why or why not?
5. The Reporter records one agreed answer in the form.

Swap roles between Task 1 and Task 2, so everyone takes a turn as Modeller.

Getting started

You will receive the notebook file (it ends in **.ipynb**) from the GitHub repository (QR code on the slides). Save it before you start.

On a university computer (JupyterLab through Anaconda)

1. Open **Anaconda Navigator** from the university's software portal (for example, the Software Hub).
2. On the **JupyterLab** tile, click **Launch**. JupyterLab opens in your web browser.
3. In the **File Browser** on the left, go to a folder you can save to, such as Documents or your H: drive. Upload the notebook with the upward-arrow button at the top of the File Browser, or drag the file onto it.
4. Double-click the notebook to open it. If you are asked to choose a kernel, choose **Python 3 (ipykernel)**, or the newest Anaconda option with Python 3.11 or later.
5. Click the first code cell and press **Shift + Enter**. The first time, it installs PyPSA, which takes one or two minutes. Wait until it prints “PyPSA … is ready.”

On your own laptop (Google Colab)

1. Go to colab.research.google.com and sign in with a Google account.
2. Choose **File > Upload notebook** and select the notebook file.
3. If Colab warns that the notebook was not written by Google, click **Run anyway**.
4. Click the first code cell and press **Shift + Enter**, then wait for "PyPSA … is ready."

The buttons you need

To do this	JupyterLab	Google Colab
Run the selected cell and move to the next	Shift + Enter, or the ▶ button	Shift + Enter, or the ▶ button at the left of the cell
Run the whole notebook	Run > Run All Cells	Runtime > Run all
Run everything above the cell you are in	Run > Run All Above Selected Cell	Runtime > Run before
Run from the cell you are in to the end	Run > Run Selected Cell and All Below	Runtime > Run after
Start again from a clean state	Kernel > Restart Kernel and Run All Cells	Runtime > Restart session and run all
See the saved figures and tables	File Browser, folder icon on the left	Files, folder icon on the left

Points to note:

- A cell is finished when a number appears in its brackets, for example **[12]**. **[*]** means it is still running.
- **Run all** stops at the battery blanks in section 4.4 until you have filled them in. That is expected.
- Grey cells are code; white cells are explanation. Read the white cell above each code cell before running it.
- Every figure and table is saved automatically in the **outputs** folder, and the last cell collects them in one zip file.

Task 1: a home with solar and a battery

A home in Nottingham has 4 kWp of solar panels and uses 2,500 kWh of electricity a year, a typical UK figure. How much of its demand can it meet itself, and what does a battery add? Time: 15 minutes, after the 5-minute set-up.

Step	Notebook section	What you do	Form
1	3	Run sections 1 to 3 from the top. Look at Figures 1 and 2: when is there spare solar?	
2	4.2	Run the baseline. Note the self-sufficiency rate	

Step	Notebook section	What you do	Form
		without a battery.	
3	4.3	Fill in the three blanks in the battery rule. Hints are under the cell. **Run the cell after every change.**	
4	4.4	Run the four-hour test. All four lines must say PASS.	
5	4.5	**Predict** the self-sufficiency rate with a 5 kWh battery, then run the cell.	Question 2, before running
6	4.6 to 4.7	Run the realistic battery and Figures 3 and 4. Why is the result lower than the ideal one?	
7	4.8	Type your group number as `MY_GROUP` and run. Agree a prediction first.	Questions 3 and 4

Groups that finish early may attempt the stretch task in section 4.9.

Task 2: the merit order and a carbon price

A simple power system has wind, nuclear, coal and gas. PyPSA finds the cheapest way to meet demand in every hour of one day. What happens when every tonne of CO2 costs 80 GBP? Time: 15 minutes. Swap roles before you start.

Step	Notebook section	What you do	Form
1	5.2 to 5.3	Look at the generator table, then run the model functions.	
2	5.4	**Predict** what a carbon price of 80 does to the peak price and to emissions.	Question 5, before running
3	5.4	Run with `CARBON_PRICE = 0`, then change it to 80 and run again.	Questions 6 and 7
4	5.5	Run Figures 6 and 7. Which power station sets the price at the peak?	
5	5.6	Run your group's scenario. Agree a prediction first.	Questions 8 and 9
6	End	Agree one sentence explaining one of your scenario results, then submit.	Question 10

Groups that finish early may attempt the stretch task in section 5.7.

Key terms in plain English

Term	Meaning

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kW and kWh	kW is power, how fast energy flows. kWh is energy, the amount used. A 2 kW heater running for 3 hours uses 6 kWh.
kWp (kilowatt-peak)	The size of a solar system: its output in ideal test conditions. A 4 kWp system rarely produces 4 kW in the UK.
PVGIS	A free European Commission service that estimates hourly solar output for any location from satellite weather data.
Self-sufficiency rate	The share of the home's yearly demand met by its own solar, directly or through the battery. 60 % means it buys only 40 % of its electricity from the grid.
Self-consumption rate	The share of the solar generation the home uses itself rather than exporting to the grid.
State of charge	How much energy is stored in the battery at a given moment, in kWh.
Round-trip efficiency	Energy you get back from a battery divided by the energy you put in. At 90 %, storing 10 kWh gives back 9 kWh.
Idealised and realistic battery	The idealised battery loses no energy and has no power limit, so it gives an upper bound. The realistic one has losses and a 3 kW limit.
Merit order	Power stations ranked from cheapest to most expensive to run. The cheapest run first.
SRMC (short-run marginal cost)	The cost of producing one more MWh: mainly fuel, plus the carbon cost.
Clearing price	The price of electricity in an hour, set by the most expensive station that has to run to meet demand (the marginal unit).
Carbon price	A charge per tonne of CO2 emitted, in GBP/tCO2. It raises the running cost of coal more than gas, because coal emits more per MWh.
Direct CO2 emissions	CO2 from burning fuel in the power station, in tonnes (tCO2).
Optimisation	Finding the best answer, here the cheapest way to meet demand, subject to limits such as each station's capacity.

Term	Meaning
	PyPSA does this for you.

If you encounter a problem

1. Read the **last line** of the error message: it names the problem.
2. Open **"If something goes wrong"** at the start of Task 1 in the notebook. It lists the common errors and their fixes.
3. For the battery blanks, open the hints under the cell in section 4.3, one at a time.
4. Look up the topic in the **Learning resources** document (the first table maps each notebook section to a help page).
5. If time is short and the battery blanks still produce an error: Click in the first code cell of section 4.6 and use **Run > Run Selected Cell and All Below** (JupyterLab), or run that cell and then **Runtime > Run after** (Colab). Everything from section 4.6 onwards works without the blanks.
6. If the problem persists, raise your hand and the instructor will assist.

The session closes with a short quiz and feedback form, accessed through the QR code on the slides.